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Track: Data Analytics
Project Phase: Week 6 Output

1. Week 5 to Week 6 Transition

In Week 5, the initial Exploratory Data Analysis (EDA) identified a critical ~48.5% no-show rate heavily driven by a 34-day average booking lead time. The objective for Week 6 was to conduct advanced multivariate analysis to identify specific demographic intersections contributing to this rate, validate logistical baselines, and execute cross-track integration to improve predictive modeling.

2. Advanced Analytical Findings (Page 2 Insights)

Multivariate Demographic Vulnerabilities: Analysis of no-shows categorized by age group and reminder channel reveals that the "None" reminder category is a primary driver of missed appointments across multiple age brackets. A one-size-fits-all communication strategy is failing to engage specific patient segments.

Validation of Logistical Factors: A comparative analysis of average wait times and clinic distances proves these are not primary drivers of no-shows. Patients who attended traveled an average of 9.68 km and waited 24.29 minutes. Patients who no-showed had nearly identical baselines (10.52 km and 24.20 minutes). Clinic resources should remain focused on lead-time interventions rather than waiting room optimization.

3. Cross-Track Integration (Data Science)

To support the broader HealthConnect project, the Data Analytics track prepared a formal feature-recommendation memo for the Data Science track. The integration objective was to ensure the baseline classification model accurately reflects operational realities. It was formally recommended that `booking_lead_days`, `reminder_channel`, and `age_group` be heavily weighted as primary predictive features.

4. Week 7 Testing Requirements

Based on the advanced findings, the Data Analytics track recommends the clinic implement an A/B test in Week 7: mandate WhatsApp confirmations for all appointments booked more than 25 days in advance for the highest-risk age brackets to measure the immediate impact on attendance rates.